

Fundamentals of Telecommunication Systems (2304211)

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	-	-	-
I	Introduction to Telecommunication	• Evolution of telecommunications • Importance and applications in various industries • Overview of voice, video, and data communication	6	CO1
II	Basics of Signals and Transmission	• Analog vs. Digital signals • Bandwidth and Frequency Spectrum • Transmission media: Wired (Copper, Fiber-optic) and Wireless	6	CO2
III	Telecommunication Networks	• Types of networks: PSTN, Cellular, Satellite, Internet • Network topologies: Star, Mesh, Ring, Hybrid • Overview of Switching and Routing	7	CO3
IV	Telecommunication Devices and Standards	• Telephone, Mobile Phones, Modems • Telecommunication standards (ITU, IEEE, 3GPP) • Basics of 2G, 3G, 4G, and 5G	6	CO4
V	Digital Communication Techniques	• Basic Modulation techniques (AM, FM, PM) • Multiplexing: FDM, TDM • Error detection and correction techniques	7	CO5
VI	Emerging Trends in Telecommunication	• IoT and Smart Communication • 5G, VoIP, and Cloud-based Communication • Cybersecurity and Data Privacy in Telecom	7	CO6

Course Outcomes

1. Describe the evolution and importance of telecommunication systems across various domains.
2. Differentiate between analog and digital signals and identify suitable transmission media.
3. Illustrate the structure and operation of different telecommunication networks and switching methods.
4. Summarize the functionality of major telecommunication devices and associated standards.
5. Explain basic digital communication techniques such as modulation, multiplexing, and error correction.
6. Discuss recent developments in telecommunications, including IoT, 5G, and cloud communication, and relate them to real-world applications.

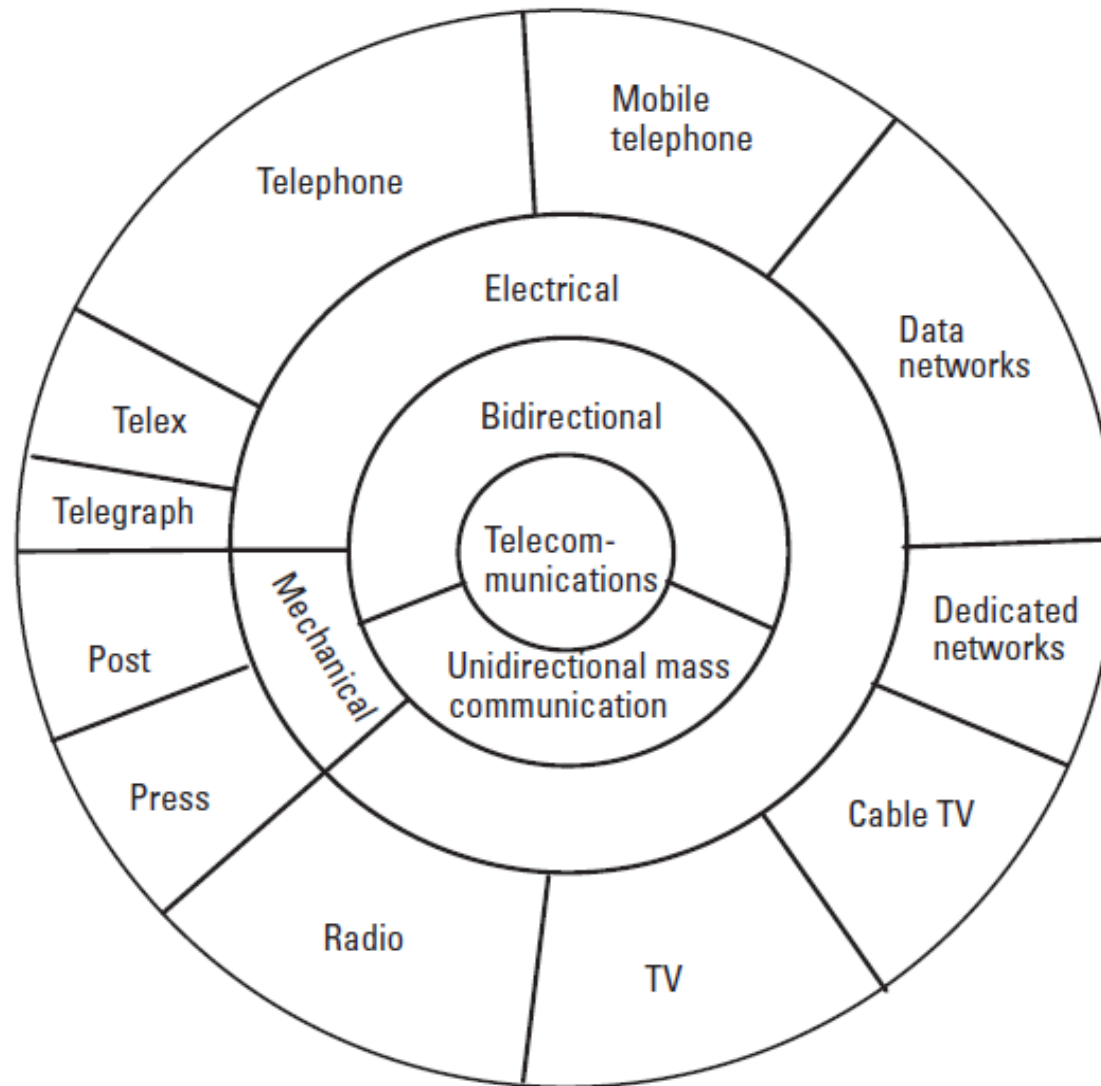
What is telecommunications?

- “tele” – Greek for *distant*
- “communicatio” – Italian for *connection*
- Telecommunication
 - distant connection or
 - transfer of **meaningful** information from one location to another
- Today it means:
 - “high tech” methods of information transfer
 - Voice
 - Video
 - Data

Introduction

- Telecommunication includes:
 1. *Mechanical communication* and
 2. *Electrical communication*.
 - This is why many authorities such as the national post, telegraph, and telephone (PTT) companies are involved in telecommunications using both forms.
 - The share of mechanical telecommunications such as conventional mail and press is expected to decrease, whereas electrical, especially bidirectional, communication will increase and take the major share of telecommunications in the future.
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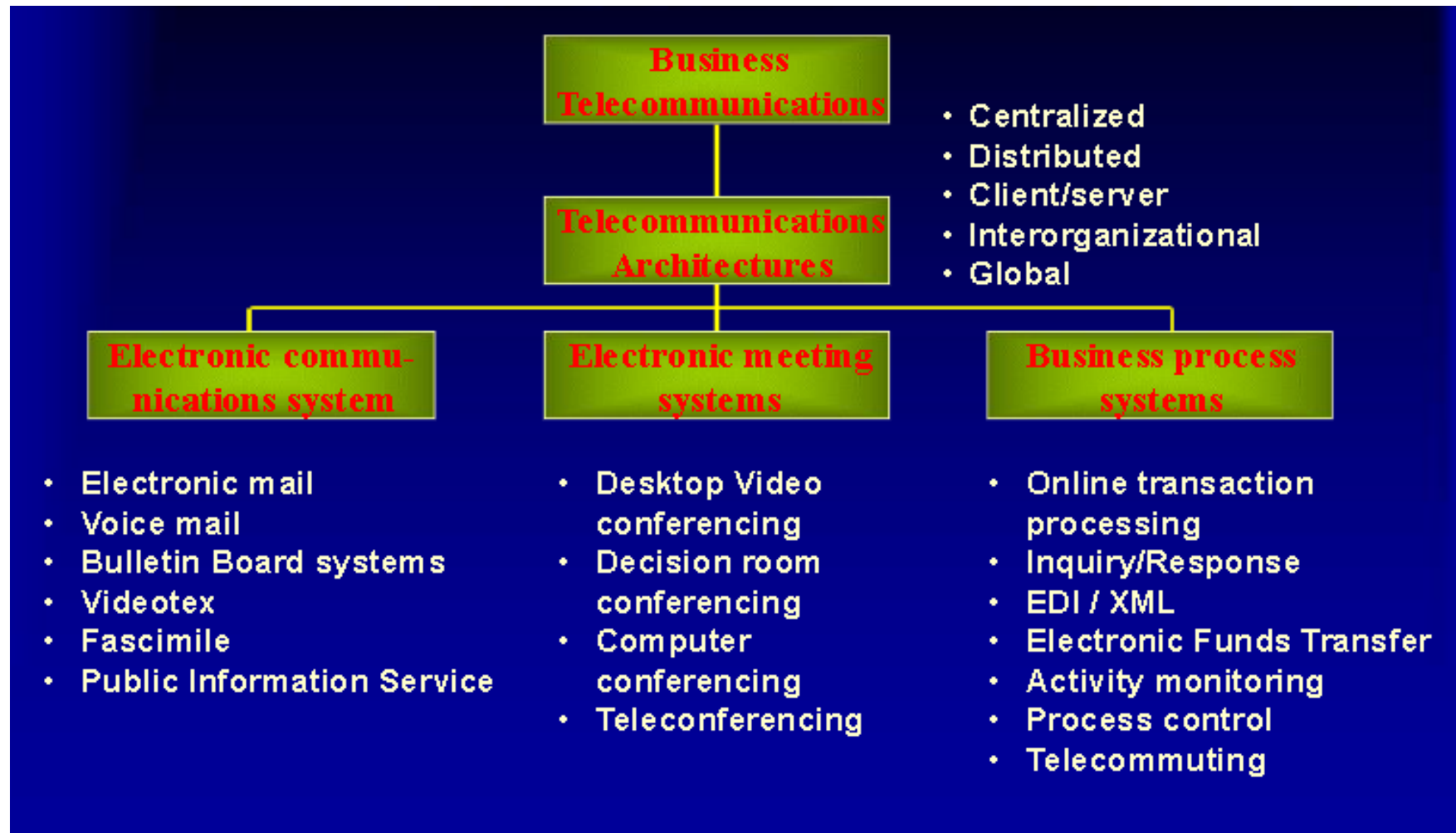
Telecommunications



Applications

- Telecommunications plays an essential role on many areas of everyday life such as:
 - ✓ Banking, automatic teller machines, tele banking
 - ✓ Aviation, booking of tickets
 - ✓ Sales, wholesale and order handling
 - ✓ Credit card payments at gasoline stations
 - ✓ Booking of hotel rooms by travel agencies
 - ✓ Material purchasing by industry
 - ✓ Government operations, such as taxation
 - Telecommunications networks make up the most complicated networks in the world.
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Applications



Applications in Various Industries

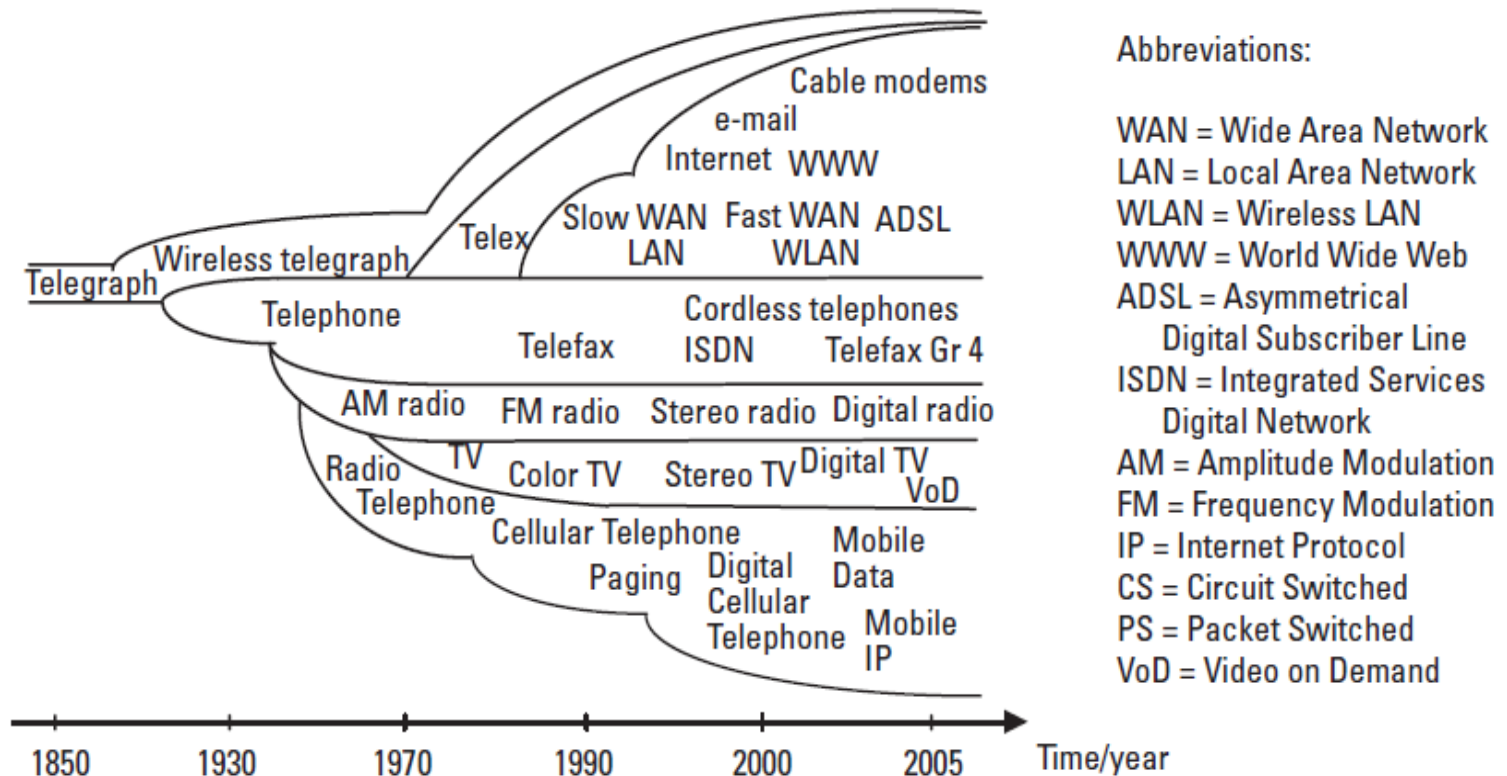
- **Banking & Finance:** Facilitates instant transactions, mobile banking, secure data exchange, and financial technology services.
 - **Healthcare:** Enables telemedicine, remote patient monitoring, and rapid sharing of medical data.
 - **Manufacturing & Logistics:** Powers IoT devices, smart metering, supply chain tracking, and real-time operational monitoring.
 - **Retail & E-commerce:** Facilitates online transactions, digital marketing, and managing supply chain visibility.
 - **Education:** Supports e-learning platforms, video conferencing for classes, and remote collaboration for students.
 - **Government & Public Safety:** Critical for emergency services, national security, and managing communication infrastructure.
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Telecommunication Importance

- **Global Connectivity & Collaboration:** Enables instant communication via video conferencing, VoIP, and messaging, facilitating 24/7 global business operations.
 - **Operational Efficiency:** Powers automation and digital processes, reducing human error and boosting productivity.
 - **Data Mobility & Cloud Integration:** Provides high-speed internet and secure,, reliable connectivity for remote workforce data access.
 - **Customer Relations:** Facilitates improved, rapid customer engagement through multichannel support (chat, voice, social media).
 - **Business Continuity & Security:** Essential for disaster recovery, secure data transmission, and protecting against cyber threats.
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Historical Perspective

- Present day telecommunication systems and services has gone through various development stages.



Development of Telecommunication Systems and Services

Timeline of Modern Telecom

- 1837– **Samuel Morse** invents **telegraph** (demonstrated in 1844)
- 1850 – **Telegraph cables** cross English Channel
- 1858 – First trans-Atlantic telegraph cable laid
- 1876 – **Alexander Graham Bell** invents **telephone**
- 1888 – **Heinrich Hertz** discovers **electromagnetic waves**
- 1895 – **Marconi** invents **wireless telegraph**
- 1901 – Marconi sends first trans-Atlantic wireless telegraph
- 1906 –The first **public radio voice broadcast**
- 1919 – **First licensed radio station** in North America
- 1924 – **First radio airing** of Stanley **Cup game**
- 1927 – **First radio** trans-Atlantic commercial **phone calls**

Timeline of Modern Telecom (2)

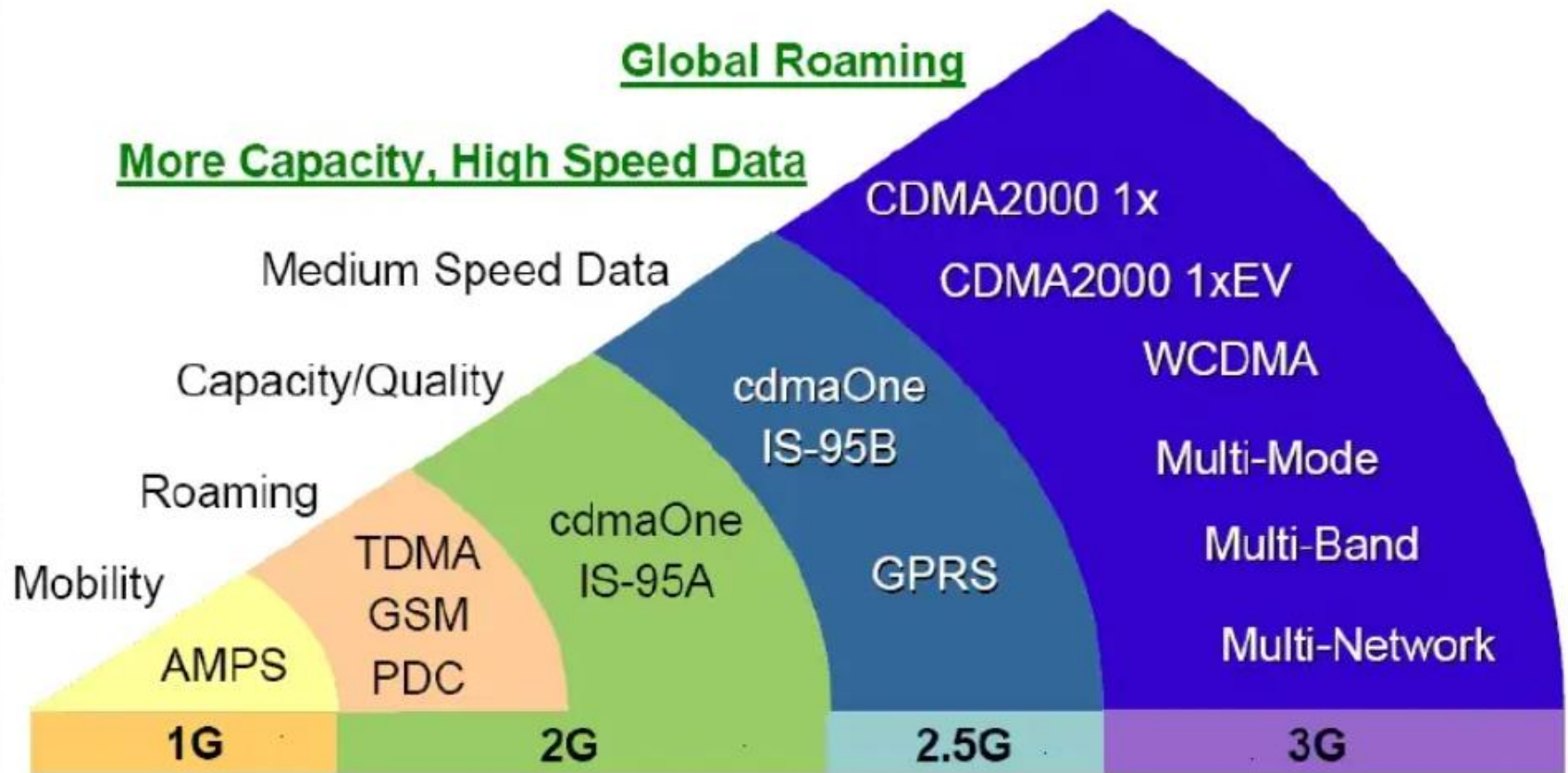
- 1939 – Electronic computer developed
- 1941 – Marriage of computer and communications
 - telegraph code punched on paper tape read by computer
- 1947/1948 – Transistor invented at Bell Labs
- 1950 – Time division multiplexing developed
- 1956 – First trans-Atlantic phone cable
- 1960 – Laser developed
- 1961 – Integrated circuit developed
- 1962 – Telstar I launched (first communication satellite)
- 1969 – Defence Advanced Research Projects Agency (DARPA) funds Advanced Research Projects Agency Network “ARPANET”
- 1970 – First optical fibre developed

Timeline of Modern Telecom (3)

- 1975 – First digital telephone switch
- 1982 – “Internet” used to describe successor to ARPANET
- 1989 – First SONET-standard optical fibre products released
- 1990 – World Wide Web becomes part of the Internet
- Today:
 - 1 billion telephones in over 200 countries
 - ~15 Billion microprocessors on the planet (6 Billion humans).
 - Telegraphy, telephony, data, television, finance, etc integrated into global telecom system
- → Telecommunications is still a growing industry !

Table 1.1 Generations of communication technology

Sl. no	Year	Generation	Key features	Limitations
1	1980s	1G	• Voice signals • Analog cellular phones • NMT, AMPS	Low security
2	1990s	2G	• Text messages • Voice signal • Digital signal • GSM, TDMA, CDMA, GPRS, EDGE	Low internet support
3	1998s	3G	• Voice signals • Data signals • Video signals • Wireless and fixed internet access • W-CDMA, UMTS	Low high-speed internet support
4	2008s	4G	• Higher data rate • Interoperability protocol • Mobile IP	Does not help with the connected devices are more than 50 billion
5	2019s	5G	• Increased connectivity • IoT • Low latency • Greater transmission speed	NMT (Nordic Mobile Telephone) AMPS (Advanced Mobile Phone System) GSM (Global System for Mobile Communications) TDMA (Time Division Multiple Access) CDMA (Code Division Multiple Access) GPRS (General Packet Radio Service) EDGE (Enhanced Data rates for GSM Evolution) UMTS (Universal Mobile Telecommunications System)



Advantages and Disadvantages



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- A complete, single telecommunications circuit consists of two stations, each equipped with a transmitter and a receiver.
 - The transmitter and receiver at any station may be combined into a single device called a transceiver.
 - The medium of signal transmission can be electrical wire or cable (also known as "copper"), optical fiber or electromagnetic fields.
 - The free-space transmission and reception of data by means of electromagnetic fields is called wireless.
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- **Terminals**

- ⇒ terminal, office equipment , telephones , ...

- **Telecommunications processors**

- ⇒ modems, multiplexers, front-end processors, ...

- **Telecommunications channels and media**

- ⇒ copper wires, coaxial cables, fiber optic cables, satellites, ...

- **Computers**

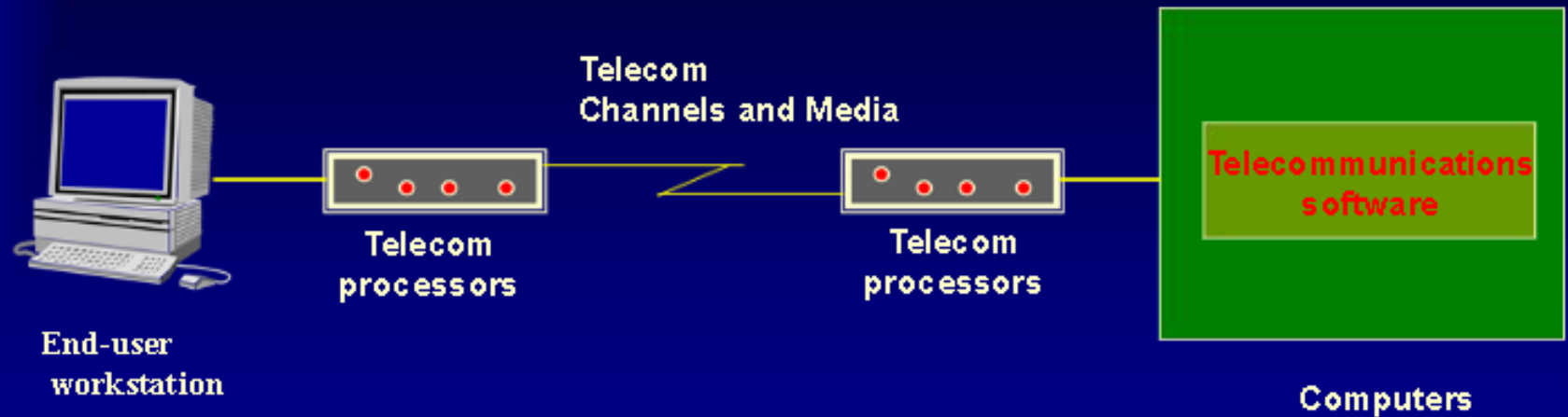
- ⇒ host computers, front-end computers, network servers, ...

- **Telecommunications control software**

- ⇒ telecommunication monitors, network operating systems, ...

Middleware

5 components



- **Signals**

- ⇒ analog
- ⇒ digital

- **Cables**

- ⇒ Twisted-pair interference
- ⇒ Coaxial cable 20x more expensive
5.500 simultaneous phone calls
- ⇒ Fiber-optic
1 fiber 30.000 phone calls

- **Wireless**

- ⇒ microwave
- ⇒ satellites

- **Radio, Infrared, Cellular Radio, Mobile computing**

- **GPS global positioning system**

Medium	Transmission speed
Metal wire	.0012Mbps - 10 Mbps
Microwave	.256 Mbps - 100Mbps
Fiber optics	.5Mbps - 1,000Mbps